

MNYMO Girls
August 15, 2026
Individual

You have 40 minutes to complete 10 problems. There is no penalty for guessing. Answers are only valid if they are written in the below table.

First and last name			
p	Answer	p	Answer
1		6	
2		7	
3		8	
4		9	
5		10	

1. What is $\frac{7776}{6} + 6 \cdot 7 + 6 - 7$?
2. Daniel, Michael, and Xindi are playing a game. Initially, Daniel, Michael, and Xindi have 1000 dollars. Each day, Xindi gives Michael 5 dollars, Michael gives Daniel 10 dollars, and Daniel gives Xindi 20 dollars. After ten days, how much money does Michael have?
3. A square is drawn with side length 1. Compute the area of the region of points with distance at most 1 from some point in the square.
4. Sneha has a deck of 10 cards labeled with positive integers 1 to 10. She draws them randomly from the deck one at a time. What is the probability that she draws both the 6 and 7 before the 8?
5. Triangle ABC has $AB = 6$, $AC = 10$, and $\angle BAC = 120^\circ$. What is its inradius?
6. If $x + \frac{1}{x} = 7$, what is the remainder when $x^6 + \frac{1}{x^6}$ is divided by 1000?
7. What is the last digit of $1^1 + 2^2 + 3^3 + 4^4 + \cdots + 100^{100}$?
8. What is the sum of the 5 smallest positive integers that each have exactly 6 positive factors?
9. Angie and Sneha are playing pickleball and will arrive at a uniformly random time between 5 and 6 PM. Angie will wait for an interval of 10 minutes whereas Sneha will wait for an interval of 15 minutes. What is the probability that they will meet and play a match?
10. Let $\triangle ABC$ have sidelengths $AB = 13$, $BC = 14$, and $AC = 15$. If there is a point D on side BC such that $BD = 6$ and a point E on side AC such that $\frac{AE}{EC} = \frac{5}{6}$, let the intersection point of lines AD and BE be X . Find the area of $\triangle AXE$.